

Phishing Campaign Profiling via Unsupervised Clustering and Anomaly Mining

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1 Project Aim and Research Questions

Phishing attacks systematically adapt social engineering hooks and obfuscate structural attributes to bypass rule-based filters. The goal of this project is to build an end-to-end, data mining pipeline that profiles structures, differentiates malicious intent from benign text, and audits evasive anomalies. We address three focused research questions directly related to our four analytical themes.

1. **Thematic Discrepancy (Text Mining):** Which topics and lexical patterns uniquely separate safe business mails from malicious phishing attempts when topics are extracted and contrasted across classes?
2. **Structural Campaign Archetypes (Clustering):** Can composite structural signatures discover cohesive attack campaign types across different clustering configurations?
3. **Evasive Anomaly Mining (Outlier Detection):** How accurately can unsupervised outlier detection methods identify rare, highly evasive phishing variations compared against an interpretable, rule-based baseline and known ground-truth labels?

2 Data and Provenance

The project integrates two established open corpora: the **Enron Email Corpus** and the **Ling-Spam Corpus**, curated via a unified benchmark subset. The fundamental unit of analysis is a single email document containing *Subject*, *Body* and *Label*.

Corpus Source	Safe (0)	Malicious (1)	Total Records	Malicious Share
Enron Corpus	15,791	13,976	29,767	46.95%
Ling-Spam Corpus	2,401	458	2,859	16.02%
Total Combined	18,192	14,434	32,626	44.24%

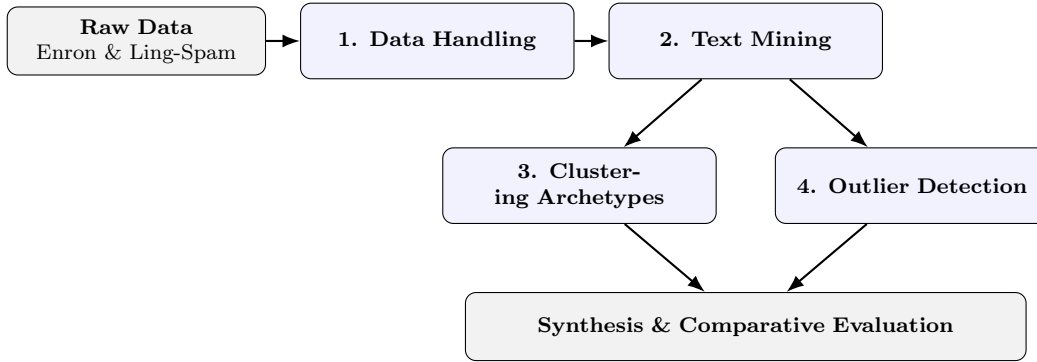
Data Quality and Sampling Strategy: Because the full merged dataset contains imbalanced representations between the two source corpora, we implement stratified downsampling for computationally intensive operations. We will also audit the safe subset to check whether benign emails are misflagged, using those cases to refine our feature set.

3 Selection of Data Mining Themes

- **Data Handling:** HTML stripping, missing value imputation. As the subject and body features are long and consists of more stuff than just pure text we have to clean the data for the future steps.
- **Text Mining:** Rather than pooling all records indiscriminately, we segment the corpus into safe and malicious subsets to contrast frequent terms and representations. We run comparative topic modeling to map safe discourse against malicious campaign vectors. This could also be seen as we are creating features from the body and subject by classifying text segments (e.g. links).
- **Clustering:** Rather than treating clustering as a single global operation, we evaluate three modular sub-datasets across three algorithms and three distance metrics:

1. *Malicious Subset*: Unsupervised discovery of cohesive attack archetypes (e.g., credential harvesting, urgent financial fraud).
 2. *Safe Subset*: Profiling benign communication structures to identify high-risk false-positive archetypes (e.g., automated IT password resets, system alerts).
 3. *Combined Dataset*: Assessing whether unsupervised topological boundaries naturally align with ground-truth labels, exposing noisy or mislabeled records.
- **Outlier Detection**: We establish an operational baseline, because 44.24% of records are malicious, an outlier is not defined simply as any phishing mail, but as a record deviating heavily from normal behavioral distributions. The baseline will be defined after gaining deeper understanding of the dataset.

4 Pipeline Diagram



5 Expected Evaluation Approach

Supervised / Ground-Truth Evaluation

- **Outlier Detection**: Evaluate anomaly detection performance against ground-truth labels using metrics such as Precision, Recall, and PR-AUC.
- **Cluster Alignment**: Compare natural cluster boundaries on the combined dataset against binary labels to test class separation.
- **False Positive Auditing**: Inspect benign emails that get flagged by models to discover label inconsistencies and refine features.

Unsupervised Evaluation

- **Data Handling**: Track record counts, deduplication rates, and missingness before and after preprocessing.
- **Text Mining**: Measure topic coherence and evaluate vocabulary overlap between safe and malicious emails.
- **Clustering Quality**: Score runs across internal validation measures and inspect top terms per cluster.
- **Outlier Stability**: Measure score correlations and ranking agreement across different anomaly detection methods.

6 Risks and Mitigation

- **Risk 1: High Sparsity in Text Feature Spaces**. High vocabulary dimensionality degrades distance-based clustering.
Mitigation: Combine dense structural features with low-dimensional topic mixture weights rather than bare high-dimensional matrices.
- **Risk 2: Computational Overhead**. Memory scaling over 32k records makes all evaluation runs intractable.
Mitigation: Draw reproducible, stratified subsets for systematic hyperparameter grid runs, preserving the global dataset for pipeline validation.

- **Risk 3: Label Noise in Historical emails.** Older email datasets often contain benign emails labeled as malicious or vice versa.
Mitigation: Treat campaign identification primarily as unsupervised pattern discovery; evaluate outlier flags via error audits rather than blindly treating ground-truth labels as absolute.